

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{xy} & T_{yy} & T_{yz} \\ T_{xz} & T_{yz} & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

$$T_{xy} = \mathbb{T}_{xy}^{(1)}$$

$$T_{xz} = \mathbb{T}_{xz}^{(1)}$$

$$T_{yy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$

$$T_{yz} = \mathbb{T}_{yz}^{(1)}$$

$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & T_{xz} \\ -T_{xy} & 0 & T_{yz} \\ -T_{xz} & -T_{yz} & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}_z^{(1)}$$

$$T_{xz} = -\mathbb{M}_y^{(1)}$$

$$T_{yz} = \mathbb{M}_x^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & T_{xxyz} & T_{xxzx} & T_{xxxy} \\ T_{xxyy} & T_{yyyy} & T_{yyzz} & T_{yyyz} & T_{yyzx} & T_{yyxy} \\ T_{xxzz} & T_{yyzz} & T_{zzzz} & T_{zzyz} & T_{zzzx} & T_{zzxy} \\ T_{xxyz} & T_{yyyz} & T_{zzyz} & T_{yzyz} & T_{yzzx} & T_{yzxy} \\ T_{xxzx} & T_{yyzx} & T_{zzzx} & T_{yzzx} & T_{zxzx} & T_{zxxy} \\ T_{xxxy} & T_{yyxy} & T_{zzxy} & T_{yzxy} & T_{zxxy} & T_{xyxy} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$

$$T_{xyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)}$$

$$T_{xxzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

$$T_{xyyz} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)}$$

$$T_{xxzx} = -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{xxxy} = \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$

$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$

$$T_{yyyz} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$

$$T_{yyzx} = \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{yyxy} = -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$

$$T_{zzyz} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$

$$T_{zzzx} = \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{zzxy} = \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{yzyz} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$

$$T_{yzzx} = \mathbb{T}_{xy}^{(2)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{yzxy} = \mathbb{T}_{xz}^{(2)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{zxxz} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$

$$T_{zxxxy} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)}$$

$$T_{xyxy} = \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xyyy} & T_{xxzz} & T_{xyyz} & T_{xxzx} & T_{xxxy} \\ -T_{xyyy} & 0 & T_{yyzz} & T_{yyyz} & T_{yyzx} & T_{yyxy} \\ -T_{xxzz} & -T_{yyzz} & 0 & T_{zzyz} & T_{zzzx} & T_{zzxy} \\ -T_{xyyz} & -T_{yyyz} & -T_{zzyz} & 0 & T_{yzzx} & T_{yzxy} \\ -T_{xxzx} & -T_{yyzx} & -T_{zzzx} & -T_{yzzx} & 0 & T_{zxxxy} \\ -T_{xxxy} & -T_{yyxy} & -T_{zzxy} & -T_{yzxy} & -T_{zxxxy} & 0 \end{bmatrix}$$

$$T_{xyyy} = 4\mathbb{M}_{xyz}^{(1)} - 2\mathbb{T}_v^{(3)}$$

$$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(3)}$$

$$T_{xyyz} = -4\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{xxzx} = -2\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{xxxy} = 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{yyzz} = 4\mathbb{M}_{xyz}^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(3)}$$

$$T_{yyyz} = 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{yyzx} = -4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{yyxy} = -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{zzyz} = -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{zzzx} = 2\mathbb{M}_y^{(1)} - 2\mathbb{M}_y^{\alpha(1)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{zzxy} = -4\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{yzzx} = -\mathbb{M}_z^{(1)} - 4\mathbb{M}_z^{\alpha(1)}$$

$$T_{yzxy} = \mathbb{M}_y^{(1)} + 4\mathbb{M}_y^{\alpha(1)}$$

$$T_{zxxxy} = -\mathbb{M}_x^{(1)} - 4\mathbb{M}_x^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & T_{yzzx} & T_{yzxy} \\ T_{yzzx} & T_{zxzx} & T_{zxxy} \\ T_{yzxy} & T_{zxxy} & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(4)}$$

$$T_{yzzx} = 2\mathbb{T}_{xy}^{(4)}$$

$$T_{yzxy} = 2\mathbb{T}_{xz}^{(4)}$$

$$T_{zxzx} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 2\mathbb{T}_v^{(4)}$$

$$T_{zxxy} = 2\mathbb{T}_{yz}^{(4)}$$

$$T_{xyxy} = \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzzx} & T_{yzxy} \\ -T_{yzzx} & 0 & T_{zxxy} \\ -T_{yzxy} & -T_{zxxy} & 0 \end{bmatrix}$$

$$T_{yzzx} = -\mathbb{M}_z^{(2)}$$

$$T_{yzxy} = \mathbb{M}_y^{(2)}$$

$$T_{zxxy} = -\mathbb{M}_x^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & T_{xxzx} & T_{xxxy} \\ T_{yyyz} & T_{yyzx} & T_{yyxy} \\ T_{zzyz} & T_{zzzx} & T_{zzxy} \\ T_{yzyz} & T_{yzzx} & T_{yzxy} \\ T_{zxyz} & T_{zxzx} & T_{zxxy} \\ T_{xyyz} & T_{xyzx} & T_{xyxy} \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)}$$

$$T_{xxzx} = \mathbb{M}_y^{(3)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{xz}^{(5)}$$

$$T_{xxxy} = \mathbb{M}_z^{(3)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} - 2\mathbb{T}_{xy}^{(5)}$$

$$T_{yyyz} = \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{yyzx} = \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)}$$

$$T_{yyxy} = \mathbb{M}_z^{(3)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{xy}^{(5)}$$

$$T_{zzyz} = \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzzx} = \mathbb{M}_y^{(3)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} - 2\mathbb{T}_{xz}^{(5)}$$

$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(5)}$$

$$T_{yzzx} = \mathbb{M}_z^{(4)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{xy}^{(5)}$$

$$T_{yzxy} = \mathbb{M}_y^{(4)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(5)}$$

$$T_{zxyz} = \mathbb{M}_z^{(4)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{xy}^{(5)}$$

$$T_{zxzx} = \mathbb{M}_{xyz}^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(5)}$$

$$T_{zxxy} = \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{yz}^{(5)}$$

$$T_{xyyz} = \mathbb{M}_y^{(4)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{xz}^{(5)}$$

$$T_{xyzx} = \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{yz}^{(5)}$$

$$T_{xyxy} = \mathbb{M}_{xyz}^{(2)} + 2\mathbb{T}_v^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & T_{yzyz} & T_{yzzx} & T_{yzxy} \\ T_{zxxx} & T_{zxyy} & T_{zxzz} & T_{zxyz} & T_{zxzx} & T_{zxxxy} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & T_{xyyz} & T_{xyzx} & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned} T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\ T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\ T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\ T_{yzyz} &= \mathbb{M}_{xyz}^{(3)} - 3\mathbb{T}_u^{(6)} - \mathbb{T}_v^{(6)} \\ T_{yzzx} &= \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + \mathbb{T}_{xy}^{(6)} \\ T_{yzxy} &= \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} - \mathbb{T}_{xz}^{(6)} \\ T_{zxxx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\ T_{zxyy} &= \mathbb{M}_y^{(5)} + 2\mathbb{M}_y^{(6)} + 2\mathbb{M}_y^{\alpha(3)} \\ T_{zxzz} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\ T_{zxyz} &= \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - \mathbb{T}_{xy}^{(6)} \\ T_{zxzx} &= \mathbb{M}_{xyz}^{(3)} + 3\mathbb{T}_u^{(6)} - \mathbb{T}_v^{(6)} \\ T_{zxxxy} &= \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + \mathbb{T}_{yz}^{(6)} \\ T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)} \\ T_{xyyy} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)} \\ T_{xyzz} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)} \\ T_{xyyz} &= \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} + \mathbb{T}_{xz}^{(6)} \\ T_{xyzx} &= \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - \mathbb{T}_{yz}^{(6)} \\ T_{xyxy} &= \mathbb{M}_{xyz}^{(3)} + 2\mathbb{T}_v^{(6)} \end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & T_{xxyz} & T_{xxzx} & T_{xxxy} & T_{xxzy} & T_{xxxz} & T_{xxyx} \\ T_{yyxx} & T_{yyyy} & T_{yyzz} & T_{yyyz} & T_{yyzx} & T_{yyxy} & T_{yyzy} & T_{yyxz} & T_{yyyx} \\ T_{zzxx} & T_{zzyy} & T_{zzzz} & T_{zzyz} & T_{zzzx} & T_{zzxy} & T_{zzzy} & T_{zzxz} & T_{zzyx} \\ T_{yzxx} & T_{yzyy} & T_{yzzz} & T_{yzyz} & T_{yzzx} & T_{yzxy} & T_{yzzy} & T_{yzxz} & T_{yzyx} \\ T_{zxxx} & T_{zxyy} & T_{zxzz} & T_{zxyz} & T_{zxzx} & T_{zxxy} & T_{zxzy} & T_{zxxz} & T_{zxyx} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & T_{xyyz} & T_{xyzx} & T_{xyxy} & T_{xyzy} & T_{xyxz} & T_{xyyx} \end{bmatrix}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & T_{yzyz} & T_{yzzx} & T_{yzxy} & T_{yzzy} & T_{yzxz} & T_{yzyx} \\ T_{zxxx} & T_{zxyy} & T_{zxzz} & T_{zxyz} & T_{zxzx} & T_{zxxy} & T_{zxzy} & T_{zxxx} & T_{zxyx} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & T_{xyyz} & T_{xyzx} & T_{xyxy} & T_{xyzy} & T_{xyxz} & T_{xyyx} \end{bmatrix}$$

$$\begin{aligned} T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\ T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\ T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\ T_{yzyz} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} + 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\ T_{yzzx} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\ T_{yzxy} &= \mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\ T_{yzzy} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} - 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\ T_{yzxz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\ T_{yzyx} &= -\mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\ T_{zxxx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\ T_{zxyy} &= \mathbb{M}_y^{(5)} + 2\mathbb{M}_y^{(6)} + 2\mathbb{M}_y^{\alpha(3)} \\ T_{zxzz} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\ T_{zxyz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\ T_{zxxz} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} + 3\mathbb{T}_u^{(6)} - 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\ T_{zxxxy} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\ T_{zxzy} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\ T_{zxxz} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} + 3\mathbb{T}_u^{(6)} + 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)} \\ T_{zxyx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\ T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)} \\ T_{xyyy} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)} \\ T_{xyzz} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)} \\ T_{xyyz} &= -\mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \\ T_{xyzx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\ T_{xyxy} &= \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(6)} \\ T_{xyzy} &= \mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \\ T_{xyxz} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\ T_{xyyx} &= \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} - 4\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(6)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & T_{xxxx} & T_{xxxx} \\ T_{yyyy} & T_{yyyy} & T_{yyyy} \\ T_{zzzz} & T_{zzzz} & T_{zzzz} \\ T_{yyyz} & T_{yyyz} & T_{yyyz} \\ T_{zzxx} & T_{zzxx} & T_{zzxx} \\ T_{xxyy} & T_{xxyy} & T_{xxyy} \\ T_{yyzz} & T_{yyzz} & T_{yyzz} \\ T_{zzxx} & T_{zzxx} & T_{zzxx} \\ T_{xyyy} + T_{yyzz} - T_{yzzx} - T_{zxzx} + T_{zzxx} & T_{xyyy} & T_{xyyy} \\ -\frac{T_{yyyy}}{2} + \frac{T_{yyzz}}{2} + T_{zxyy} & T_{xyyz} + \frac{T_{zzzz}}{2} - \frac{T_{zzzz}}{2} & \frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} + T_{xyyy} - T_{yyyy} + T_{zzxy} \end{bmatrix}$$

$$T_{xxxx} = 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$

$$T_{xxxy} = \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{xxxz} = -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{yyyx} = -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{yyyy} = 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$

$$T_{yyyz} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzzx} = \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{zzzy} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzzz} = 3\mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$

$$T_{yyzx} = \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{yyzy} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{zzxx} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{zzxy} = \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{zzxz} = \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{xxyx} = \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{xxyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)}$$

$$T_{xxyz} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{yzzx} = \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{yzzy} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

$$T_{yzzz} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{zxxx} = -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{zxxxy} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$

$$T_{zxxz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

$$T_{xyyy} = -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{xyyz} = \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)}$$